

TRUE-UIE: Two Universal Relations Unify Information Extraction Tasks

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Abstract

Information extraction (IE) encounters challenges due to the variety of schemas and objectives that differ across tasks. Recent advancements hint at the potential for universal approaches to model such tasks, referred to as Universal Information Extraction (UIE). While handling diverse tasks in one model, their generalization is limited since they are actually learning task-specific knowledge. In this study, we introduce an innovative paradigm known as TRUE-UIE, wherein all IE tasks are aligned to learn the same goals: extracting mention spans and two universal relations named NEXT and IS. During the decoding process, the NEXT relation is utilized to group related elements, while the IS relation, in conjunction with structured language prompts, undertakes the role of type recognition. Additionally, we consider the sequential dependency of tokens during span extraction, an aspect often overlooked in prevalent models. Our empirical experiments indicate that TRUE-UIE achieves state-of-the-art performance on established benchmarks encompassing 16 datasets, spanning 7 diverse IE tasks. Further evaluations reveal that our approach effectively share knowledge between different IE tasks, showcasing significant transferability in zero-shot and few-shot scenarios.

1 Introduction

Information Extraction (IE) refers to the task of automatically extracting structured knowledge, including entities, relations, events, and sentiments, from unstructured textual data. The primary aim is to condense text into structured, machine-friendly formats, aiding downstream tasks such as question answering (Allam and Haggag, 2012) and sentiment analysis (Medhat et al., 2014).

In the era of Large Language Models (LLMs), structured knowledge enhances, validates, and

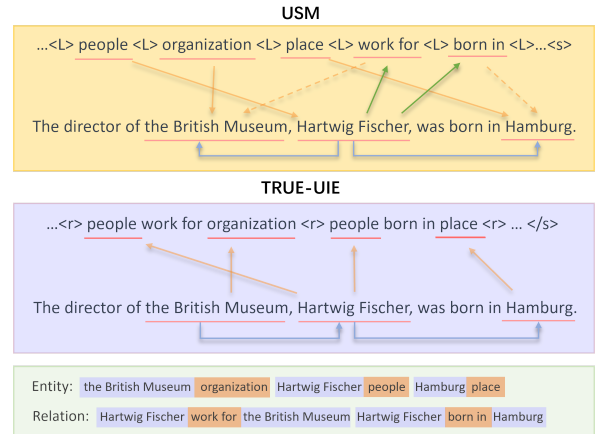


Figure 1: TRUE-UIE’s superiority over USM: unifying its framework with (1) structure language prompts and (2) only two relations, IS (yellow) and NEXT (blue), circumventing the inconsistent learning objectives encountered by USM.

grounds LLM outputs (Pan et al., 2023). Researchers are increasingly focusing on Universal Information Extraction (UIE), aiming to develop unified frameworks for various IE tasks. Two primary approaches have gained prominence: generative methods and linking-based methods. Generative methods generate a unified Structure Extraction Language to express various extraction targets (Lu et al., 2022; Cong et al., 2023). Linking-based methods, on the other hand, devise a set of directed token linking operations to break down information extraction tasks into multiple token pair labeling problems (Lou et al., 2023; Yan et al., 2023; Ping et al., 2023). Although both claim to be universal information extraction methods, We hold the belief that *a true UIE should maintain a uniform learning objective for all IE tasks, enabling comprehensive knowledge sharing*. Generative methods deviate from this criterion, generating specific structure languages for different IE tasks (Lu et al., 2022). For instance, structures generated for Named Entity Recognition tasks (NER)

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lack the use of nesting “()”, while those for relation and event extraction structures involve varying degrees of nesting “()”. Existing linking-based methods also fail to meet this criterion. Take the prominent work USM (Lou et al., 2023) as an example (Figure 1): both the dashed and solid yellow arrows are defined identically but serve different purposes in NER and RE tasks. This leads to distinct learning objectives and limited knowledge sharing. Furthermore, the relations represented by green and blue arrows are only used in the RE task and receive no training in the NER task. Similar inconsistencies are evident in other linking-based methods (Yan et al., 2023; Ping et al., 2023). Additionally, all existing UIE methods face challenges in handling complex IE tasks, like discontinuous NER and open information extraction.

In this paper, we introduce **TRUE-UIE**, **Two Universal Relations Unify Information Extraction Tasks**, a novel approach distinguishes itself from prior work by modeling all information extraction tasks as a common task, with the aim of conducting two universal relation extractions. This achievement marks a paradigm shift towards the applicability of universal model outputs, moving away from outputs tailored to specific tasks. The success of TRUE-UIE hinges on two distinct designs: (1) **Structure Language Prompt**: The structured information of schemes is preserved, and placeholders for the IS relation are left for target mentions in the text. For instance, in the task of relation extraction, we organize prompts as *<subject type> <relation type> <object type>* as shown in Figure 1, in contrast to USM which separately enumerate entity and relation types. (2) **Only two relations are employed**: IS and NEXT. The IS relation aligns spans with corresponding placeholders in the prompt. As depicted in Figure 1, the entity "Hartwig Fischer" is linked to the entity type "people" in the triplet scheme *people work for organization*, indicating that "Hartwig Fischer" is involved in a relation of type *work for* and is categorized as "people". On the other hand, the NEXT relation establishes a connection between the current span and the subsequent span within the same structural knowledge instance. For instance, "Hartwig Fischer" is linked to "Hamburg" through the NEXT relation, indicating their membership in the same triplet. Using this approach, the IS relation is utilized to identify span types, while the NEXT relation groups these spans effectively. Additionally, this method tackles the challenge of a span appearing

in several instances of the same knowledge type, a common challenge in overlapping relation extraction. (Wang et al., 2020). This is also why USM must employ the green relation in the RE task.

We conducted comprehensive experiments on 16 datasets covering 7 IE tasks, including flat NER, relation extraction, event extraction, sentiment extraction, nested NER, discontinuous NER, and open information extraction. These experiments demonstrate that TRUE-UIE surpasses both state-of-the-art task-specific and universal IE models across all datasets. Additionally, further zero-shot and few-shot experiments indicate that TRUE-UIE’s universal relations enable more effective knowledge transfer across tasks.

2 Related Work

Information Extraction (IE) is the task of extracting relevant spans or tuples of spans from plain text. There are various specific IE tasks, including Flat/Nested/Discontinuous Named Entity Recognition (Nadeau and Sekine, 2007), Relation Extraction (Nasar et al., 2021), Event Extraction (Li et al., 2022b), Sentiment Extraction (Schouten and Frasincar, 2015), and Open Information Extraction (Zhou et al., 2022). For an extended period, researchers have focused on devising task-specific and independent methods to address these diverse IE tasks. However, in recent years, the emergence of pretraining techniques has sparked considerable interest in pretraining a versatile model capable of handling multiple IE tasks. Yan et al.2021b were the first to propose a universal approach to tackling different NER tasks. Yan et al.2021a unified various aspect-based sentiment analysis tasks. Lu et al.2022 introduced UIE, which employs a Structured Extraction Language to frame all IE tasks. Building upon UIE, Cong et al.2023 incorporated meta-pretraining to enhance the model’s ability to extract complex structures. In contrast to UIE’s use of a sequence-to-sequence structure to directly generate diverse target information structures, borrowing the idea from token pair linking (Wang et al., 2020, 2021b; Yu et al., 2022), USM (Lou et al., 2023) introduces three unified token linking operations to capture the skills of structuring and conceptualizing. Similarly, UTC-IE (Yan et al., 2023) decomposes several IE tasks into token pair classification tasks, utilizing the starting and ending tokens to locate spans, and using start-to-start and end-to-end token pairs to establish relations.

UniEX (Ping et al., 2023) also uniformly dissects all extraction objectives into joint span detection, classification, and association problems through a unified extractive framework. However, existing generative (Lu et al., 2022; Paolini et al., 2021; Cong et al., 2023; Wang et al., 2023) or token pair linking methods (Lou et al., 2023; Yan et al., 2023; Ping et al., 2023; Zhu et al., 2023) still struggle to unifying all Information Extraction (IE) tasks into a single learning objective, thus maximizing knowledge sharing and generalization. In contrast, our proposed True-UIE utilizes two universal relations to harmonize all tasks.

3 Methodology

Information extraction is the process of extracting knowledge from unstructured textual sources. The primary objective of UIE is to establish a single, universal model that can handle various information extraction tasks. The challenges of current SOTA method USM encompass two main dimensions: (1) Adapting the model to address the continually evolving complexities of information extraction, particularly in contexts where discontinuities and overlapping issues emerge; (2) Enhancing the model’s generalization capabilities to ensure a broader degree of knowledge transferability and sharing across diverse tasks.

In this section, we begin by outlining the overall architecture and core principles of TRUE-UIE. Subsequently, we elucidate how TRUE-UIE addresses the aforementioned challenges. This entails two pivotal ideas: First, the introduction of a structural language prompt. By incorporating structured information from the schema into the prompt, we aim to enhance the model’s comprehension of tasks and alleviate its learning burden. Second, utilizing two universal relational edges in conjunction with the structural prompt, we manage to unify seven IE tasks, transmuting them into a unified linking task with universal scheme. This strategy seeks to maximize the potential for knowledge to be shared seamlessly across tasks. Lastly, we introduce the main mathematical formulas and training objectives involved in the model.

3.1 The Overall Architecture

As illustrated in Figure 2, TRUE-UIE creates a structural prompt (enclosed in a purple dashed line) based on the extraction demands of the task, and concatenates it with the input text. The combined

input is first passed through an encoder to obtain hidden states. These output hidden states are then processed by two fully connected layers, resulting in two distinct representations. Both representations are fed into the Semi-Matrix BiLSTM module and the Multiplicative Attention module. The operations of these two modules, shown on the right, produce presentations of spans and the corresponding relation scores. The span presentations are further used to compute the scores of spans through a fully connected layer.

3.2 Linking Scheme

Given an input text, TRUE-UIE combines the structure language prompt with the text to cater to varying extraction requirements. The adoption of this particular prompt arises from a notable distinction from previous work, where the structured information from the schema was not incorporated into the prompt. This forced the model to learn the intricate structure for each individual task. Regrettably, this knowledge could not be easily transferred across tasks, as each task possessed its unique structure.

The combined text is then input into the model, leading to the creation of a linking matrix that captures the relationships between tokens. In this framework, the IS relation aligns spans with their corresponding concept placeholders in the prompt, while the NEXT relation establishes a connection between a current span and the following span within the same instance of structural knowledge, such as within a triplet, an event, or an open fact. Next, we will provide a detailed presentation of the linking specifics for each IE task.

Relation Extraction: As illustrated in Figure 1, entity types and a relation type are amalgamated into a triplet prompt in the format of *<subject type> <relation type> <object type>*. Given that relation types often function as predicates, this design renders the prompt akin to a natural language expression, which facilitates semantic matching by the model. In cases of pure relation extraction where entity type annotations are absent, entity types default to “subject” or “object.” When two utterance spans are connected by a NEXT relation and individually link to the subject type and object type surrounding the same relation type, a triplet is ascertained. Throughout this process, both entity types and the relation type are simultaneously determined. Even when a triplet involves multiple identified entity types, this decoding method does not introduce errors. Conversely, models with naive

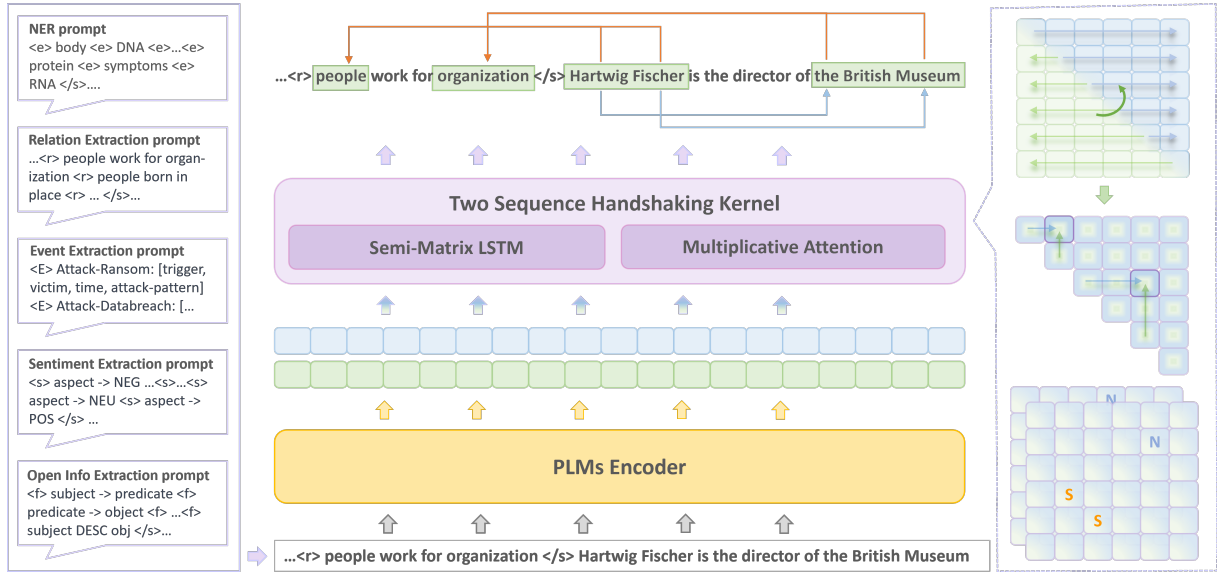


Figure 2: The overall architecture of TRUE-UIE.

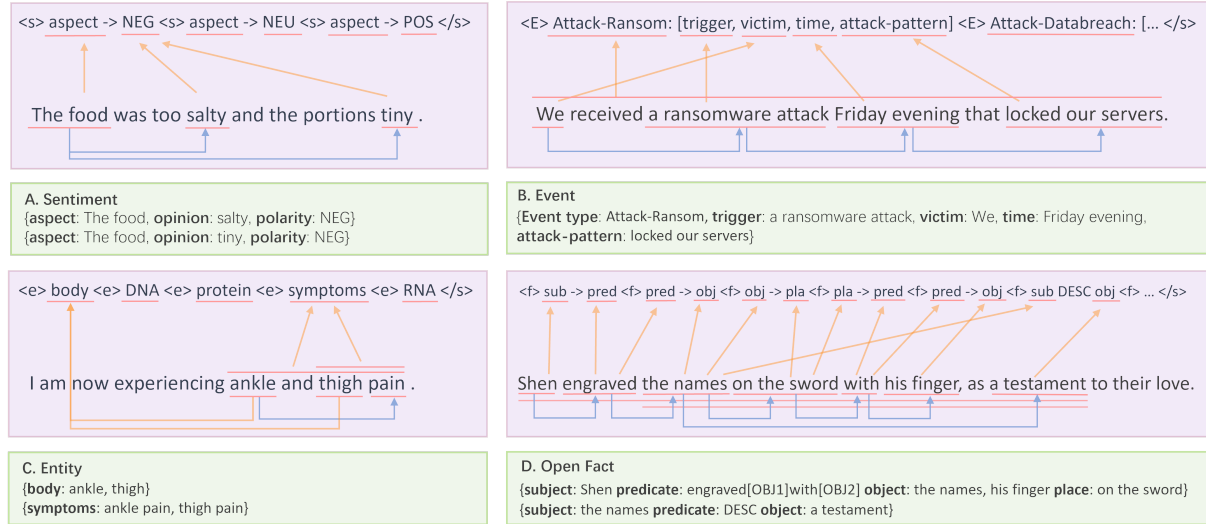


Figure 3: Unify different knowledge structures as two universal relations: IS (yellow lines) and NEXT (blue lines).

prompts struggle as they cannot discern which entity type(s) correspond to the recognized relation triplet, as they identify the entity type and relation type separately.

Sentiment Extraction: As illustrated in Figure 3.A, TRUE-UIE constructs a prompt for each sentiment type using the format *aspect* → *<polarity>*. This approach is analogous to relation extraction. When two spans are connected by a NEXT relation, and individually link to the “aspect” and the *<polarity>* surrounding the same →, a sentiment triplet is thereby determined.

Event Extraction: For representing an event, TRUE-UIE constructs a prompt using the format *<event type>*: [*argument role1*, *argument role2*,

...], where the trigger is also considered as an argument, as depicted in Figure 3.B. During the decoding process, all spans that are linked to argument roles by the IS relation are grouped according to the preceding event type. Within the entire event span (indicated by the long red line above the text), only those paths that consist of argument spans sequentially linked by the NEXT relation and extending from one boundary to the other are outputted as individual event instances. Through this decoding logic, the model can effortlessly ascertain to which event type and trigger an argument span belongs, thereby smoothly resolving the event overlapping issue, where an argument may serve different roles within different instances of the same event type.

Conversely, models employing naive prompts grapple with this overlapping problem.

Nested and Discontinuous NER: For this task, TRUE-UIE employs a prompt similar to the naive one used in previous models. However, by utilizing the relation NEXT, TRUE-UIE gains the ability to handle discontinuous entities. Specifically, TRUE-UIE examines every span linked to an entity type to determine if there exists a continuous path within it, comprised of shorter spans, stretching from one boundary to the other. If such a path is found, it is output as a discontinuous entity, and the longer span is disregarded, as illustrated by *ankle pain* in Figure 3.C. If no path is found, the span is considered as a continuous entity. Additionally, if a short span is encompassed within a longer one without a connecting path, both are recognized as entities, reflecting a nested situation. An example of this is the term *thigh*, which appears within the spans *ankle and thigh pain* and *thigh pain*, but is not part of any path. As a result, *thigh* is identified as a *body* entity based on the IS relation, *thigh pain* is recognized as a *symptom* entity, and *ankle and thigh pain* is omitted, as previously described.

Open Information Extraction: This task involves identifying common role types such as subject, predicate, object, place, time, qualifier, etc., as demonstrated in Figure 3.D. This task faces challenges such as discontinuous arguments and role overlapping (e.g., "the names" serving as both object and subject). To tackle these complexities, TRUE-UIE uses the path decoding method with long spans and NEXT relations, as previously mentioned in discontinuous NER and event extraction. It avoids linking spans to a singular role through the IS relation, as this would not resolve the overlapping issue. Instead, TRUE-UIE recognizes roles in pairs like $\langle \text{role1} \rangle \rightarrow \langle \text{role2} \rangle$, where two spans sequentially linked by NEXT and associated with role1 and role2 nearby the same $\rightarrow$ determine the roles. This ensures that every begin-to-end path within a long span is outputted as a fact instance. In situations where a predicate is missing, TRUE-UIE checks if subject and object spans are linked to predefined predicates, adding them to the fact instance if needed. An example of this can be found in the descriptive (DESC) fact in Figure 3.D.

3.3 Model Architecture

In previous linking-based UIE methods, span extraction often focuses only on the beginning and ending tokens of a span, neglecting the information

embedded within the inner tokens. This can leave valuable sequential dependencies unexploited, particularly those crucial to the extraction of spans. In contrast, TRUE-UIE explicitly utilizes all tokens within a span. By employing semi-matrix LSTM operations to efficiently embeds this information into the span features. Given a sequence of n tokens $[t_1, \dots, t_n]$, each token t_i is initially transformed into a low-dimensional contextual vector h_j utilizing a pretrained language model encoder such as BERT (Devlin et al., 2019) or RoBERTa (Liu et al., 2019). Subsequently, two distinct representations, h_j^b and h_j^e , are computed to serve as features, specifically denoting the beginning and ending tokens of span boundaries:

$$h_j^b = W_b \cdot h_j + b_b, \quad (1)$$

$$h_j^e = W_e \cdot h_j + b_e. \quad (2)$$

Herein, W_* represents a parameter matrix, and b_* is a bias vector, both of which are subject to optimization during the training process.

For both h^b and h^e , TRUE-UIE constructs two matrices B and E by repeating each vector n times, each of dimensions $n \times n$, where n is the number of tokens. Next, TRUE-UIE employs a forward LSTM to encode the upper triangular region of E and a backward LSTM to encode the lower triangular region of B . The result is two new matrices B' and E' , both of dimensions $n \times n$. In these matrices, the element $B'_{i,j}$ comprises the sequential information extending from token j to token i , while the corresponding element $E'_{i,j}$ embodies the sequential information extending from token i to token j . Subsequently, TRUE-UIE transposes B' , and the sum of B' and E' yields a new matrix, denoted as S , where only the upper triangular region is saved, and the element $S_{i,j}$ encompasses the sequential information from token i to j as well as from j to i . This structured transformation facilitates TRUE-UIE's capacity to discern intricate dependencies between the tokens, thereby aligning with the overarching objective of span extraction. The mathematical formulations for scoring a span are provided as follows:

$$S_{i,j} = BiLSTM([h_i, \dots, h_j]), \quad (3)$$

$$s_{i,j}^p = W_s \cdot S_{i,j} + b_s. \quad (4)$$

Herein, the $BiLSTM$ serves as a succinct expression for encoding the sequential information mentioned above. The score $s_{i,j}^p$ represents the output

score for the span extending from token i to token j .

Additionally, when decoding the relations between two spans, a relation (IS or NEXT) is determined to exist only if both the beginning and ending tokens of the spans share this relation. TRUE-UIE adopts a multiplicative attention operation to fuse the features of these token pairs, feeding the integrated information to relation scorers:

$$s_{i,j}^* = h_i^* \cdot h_j^{*T}, \quad (5)$$

where h^* denotes the previously described features associated with the span boundaries, as expressed in Equations 1 and 2, the asterisk (*) symbolizes either b for beginning or e for ending of a span. The score $s_{i,j}^*$ signifies the relation score between the two boundary tokens i and j .

3.4 Learning Objective

The training process encounters a class imbalance issue, where the relation IS tends to occur more frequently than NEXT across all tasks. This disproportion is particularly pronounced in NER tasks, where discontinuous entities make up a small proportion, resulting in the relative sparsity of the NEXT relationship. To address this challenge, following USM (Lou et al., 2023), we implement optimization on class imbalance loss (Su et al., 2022):

$$L = \sum_{t \in T} \log \left(1 + \sum_{(i,j) \in t^+} e^{-s_{i,j}^*} \right) \quad (6)$$

$$+ \log \left(1 + \sum_{(i,j) \in t^-} e^{s_{i,j}^*} \right) \quad (7)$$

In this part, let T denote the set of label types, where t^+ corresponds to the target class, and t^- represents the non-target class. In this context, $s_{i,j}^*$ designates the scores as defined in Equations 4 and 5, with the asterisk (*) symbol taking on the values p for a span, b for the beginning pair, and e for the ending pair.

4 Experiment

In this section, comprehensive experiments are undertaken in both the supervised setting and few-shot/zero-shot scenarios. We also provide ablation study on each component of TRUE-UIE in Appendix.

4.1 Experimental Setup

In the supervised setting, we conduct experiments across 4 information extraction tasks commonly utilized in previous research (Yan et al., 2023; Lou et al., 2023; Ping et al., 2023), including namely, flat named entity recognition, relation extraction, event extraction, and sentiment extraction. Moreover, to further substantiate TRUE-UIE’s scalability and effectiveness, we have added 3 additional tasks (nested, discontinuous named entity recognition, and open information extraction). Thus, this part of the experimentation covers seven information extraction tasks and utilizes 16 publicly available benchmark datasets only for research purposes, consistent with their intended use. The datasets employed include ACE04 (Mitchell et al., 2005), ACE05 (Walker et al., 2006); CoNLL03 (Sang and De Meulder, 2003), GENIA (Kim et al., 2003), Cadec (Karimi et al., 2015), CoNLL04 (Roth and Yih, 2004), SciERC (Luan et al., 2018), NYT (Riedel et al., 2010), CASIE (Satyapanich et al., 2020), SemEval-14/15/16 (Pontiki et al., 2014, 2015, 2016), and Saoke (Sun et al., 2018). The evaluation metrics align with those employed by Lu et al. (2022).

We primarily contrast TRUE-UIE with the previous SOTA model, USM (Lou et al., 2023), adhering to the same settings they employ for experiments. During the pretraining phase, we follow USM to use three corpus:

- D_{task} refers to Ontonotes (Pradhan et al., 2013), a widely used IE dataset. Each instance comes with a gold annotation, enabling the acquisition of in-task knowledge.
- D_{dist} represents the datasets obtained through distant supervision, wherein each instance aligns the text with Wikidata and Freebase (Cabot and Navigli, 2021; Riedel et al., 2013). Distant supervision is employed to gather large-scale training signals (Mintz et al., 2009), supplementing in-task supervised signals.
- D_{ind} denotes the indirect supervision dataset, comprising instances derived from sources outside the IE tasks. Following the USM setting, we leverage comprehension datasets from MRQA (Fisch et al., 2019) to offer a more enriched label semantic context for pre-training. Within this setting, questions are treated as labels.

Dataset	Tailored Model	UIE	UniEX	UTC-IE	USM*	USM [†]	USM ^u	TRUE*	TRUE [†]	TRUE ^u	
ACE04	P-NER	88.72	86.89	87.12	87.54	87.79	87.62	87.34	88.92	89.34	89.91
ACE05-Ent	P-NER	88.26	85.78	87.02	87.75	86.98	87.14	-	88.31	90.10	-
CoNLL03	BS	93.65	92.99	92.65	93.45	92.76	93.16	92.97	92.88	93.51	94.13
Genia	PIQN	81.77	-	76.69 ⁻	80.45	-	-	-	80.46	81.83	82.56
Cadec	W ² NER	73.21	-	-	-	-	-	-	72.06	73.25	73.83
Cadec _D	Mac	44.40	-	-	-	-	-	-	46.31	47.15	47.51
ACE05-Rel	PURE	69.40	66.06	66.06	67.79 ⁺	66.54	67.88	-	67.93	70.84	-
CoNLL04	REBEL	75.40	75.00	73.40	-	75.40	75.86	78.84	73.05	77.84	78.94
NYT	UniRel	93.70	93.54	-	-	93.96	94.07	94.01	93.98	94.33	94.83
SciERC	PFN	38.40	36.53	38.00	38.77 ⁺	37.05	37.36	37.42	37.40	38.06	38.85
ACE05-Evt _T	QE	73.60	73.36	74.08	73.44 ⁺	71.68	72.41	72.31	72.51	74.63	76.42
ACE05-Evt _A	QE	55.10	54.79	53.92	57.68 ⁺	55.37	55.83	55.57	55.21	56.41	56.81
CASIE _T	Txt2Evt	68.98	69.33	71.46	-	70.77	71.73	71.56	71.32	72.53	73.02
CASIE _A	Txt2Evt	60.37	61.30	62.91	-	63.05	63.26	63.00	62.78	63.66	63.90
14-res	GAS	72.16	74.52	74.77	-	76.35	77.26	77.29	77.11	77.82	78.13
14-lap	GAS	60.78	63.88	65.23	-	65.46	65.51	66.60	66.03	66.94	67.07
15-res	Sp-ASTE	63.27	67.15	68.58	-	68.80	69.86	-	69.92	70.78	-
16-res	Sp-ASTE	70.26	75.07	76.02	-	76.73	78.25	-	77.76	78.83	-
SAOKE	DragonIE	46.10	-	-	-	-	-	-	43.34	46.51	47.11

Table 1: The main results in the supervised setting. TRUE-UIE employs RoBERTa-large for English tasks and employs XLM-RoBERTa-large for SAOKE, as the latter needs to be trained on both Chinese and English datasets. The symbol $\star$ indicates that the model is initialized from the original pre-trained language model, $\dagger$ and u separately denote the models that were pre-trained on $D_{task,dist,ind}$ and fine-tuned on a single task and multi-task except for overlapped datasets: ACE05-Ent/Rel and 15/16-res. The symbol $+$ is used to represent results derived from models that are domain-specific or larger in size compared to RoBERTa-large. Cadec_D refers to the subset of entities that are discontinuous.

Unseen/All	10/12	7/9	6/7	8/9	7/8	8/9	4/5	12/17	Avg	Improv
D_{task}	32.1/ 33.9	2.5/ 4.3	1.6/ 2.8	10.7/ 12.2	52.4/ 53.9	45.9/ 47.4	11.2/ 12.7	14.1/ 15.4	21.3/ 23.1	+ 1.8
$D_{task,ind}$	39.8/ 41.9	14.7/ 16.2	20.6/ 22.5	24.1/ 26.1	56.2/ 57.9	44.2/ 46.1	32.9/ 34.5	44.3/ 45.9	34.6/ 36.3	+ 1.7
$D_{task,dist}$	35.4/ 38.6	21.1/ 24.2	40.6/ 43.0	27.6/ 30.3	57.0/ 60.2	49.3/ 52.1	43.7/ 46.1	44.1/ 47.3	39.8/ 42.7	+ 2.9
$D_{task,ind,dist}$	42.1/ 45.3	26.0/ 29.1	44.4/ 47.3	34.9/ 38.1	65.7/ 68.9	60.1/ 63.1	56.7/ 59.9	55.3/ 58.5	48.1/ 51.3	+ 3.2
Δ	10.0/ 11.4	23.5/ 24.8	42.7/ 44.5	24.2/ 25.9	13.3/ 15.0	14.1/ 15.7	45.5/ 47.2	41.1/ 43.1	26.8/ 28.2	-

Table 2: Comparison of zero-shot transfer performance on unseen entity label subset with different supervision signals between USM and TRUE-UIE, with two scores separated by “/”. “Unseen” indicates label types that do not appear in the pre-training dataset. “Avg” represents average scores under pretraining; “Improv” indicates average improvement against USM; Δ signifies the enhancement difference from $D_{task,ind,dist}$ to D_{task} .

Placeholder	CoNLL04	Model Size
GPT-3	18.10	175B
DEEPSTRUCT	25.80	10B
USM	25.95	356M
TRUE-UIE	27.13	374M

Table 3: Zero-shot performance on relation extraction.

In addition to USM, we also make comparisons with two other linking-based UIE models (Yan et al., 2023; Ping et al., 2023) and a Generative UIE model (Lu et al., 2022). Towards providing a thorough evaluation of TRUE-UIE’s performance relative to contemporary approaches, task-tailored

models are also in comparison: PIQN (Shen et al., 2022), W²NER (Li et al., 2022a), Mac (Wang et al., 2021b), Txt2Evt (Lu et al., 2021), PURE (Zhong and Chen, 2021), DragonIE (Yu et al., 2022), BS (Zhu and Li, 2022), P-NER (Shen et al., 2023), REBEL (Huguet Cabot and Navigli, 2021), UniRel (Tang et al., 2022), PFN (Yan et al., 2021c), QE (Wang et al., 2021a), GAS (Zhang et al., 2021), Sp-ASTE (Xu et al., 2021).

For additional details regarding the datasets, metrics, and training implementation, please consult Appendix A.

4.2 Experiments in the Supervised Setting

Table 1 presents the performance of TRUE-UIE and strong baselines. Through the observation of experimental results, we identify several advantages of the TRUE-UIE framework, setting new state-of-the-art in the field of UIE.

1) TRUE-UIE offers a universal design that facilitates seamless sharing of learned knowledge across tasks. USM’s decline in performance on several datasets after multi-task training ($USM^\dagger$ vs. USM^u) suggests that its design may hinder proper knowledge sharing across tasks, potentially leading to conflicts among them. TRUE-UIE overcomes this by transforming multi-tasks into a unified common task, demonstrating more stable growth under the same experimental settings ($TRUE^\dagger$ vs. $TRUE^u$). 2) TRUE-UIE is not merely a more universal framework but also exhibits a strong advantage in initial performance before pretraining. It surpasses other pretrained UIE methods even before pre-training. Particularly in NER tasks, where TRUE-UIE’s prompt and linking style are almost identical to USM’s design, it still significantly outperforms USM on various datasets. This improvement is attributed to the token sequential information embedded in the span features, which, apart from the prompt and linking style, is the main distinction from USM. 3) TRUE-UIE showcases the ability to tackle discontinuous and overlapping issues, a capability lacking in earlier linking-based UIE models. Although the initial performance of TRUE-UIE falls short of task-specific state-of-the-art models, after pre-training, it attains improvements of 3.11 on $Cadec_D$ and 1.01 on SAOKE, respectively. TRUE-UIE’s universal design, prioritizing overall performance across all tasks, explains why it might not excel in specific tasks without prior pre-training. 4) It is noteworthy that after multi-task fine-tuning on English datasets, TRUE-UIE demonstrates a slight improvement on SAOKE (+0.6), a Chinese dataset. This reveals TRUE-UIE’s promising ability to generalize knowledge across languages.

4.3 Experiments in the Zero-shot Setting

In zero-shot NER setting, aligned with USM, TRUE-UIE is trained using 4 different combinations of pretraining datasets and then evaluated across 8 diverse NER datasets (Liu et al., 2013; Strauss et al., 2016; Liu et al., 2021). As illustrated in Table 2, in four pre-training settings, TRUE-UIE consistently outperforms USM across all datasets,

highlighting its strong zero-shot transferability across various domains. This shows a more robust generalization capability than USM. Moreover, comparative analysis reveals a notable expansion in the performance growth gap for TRUE-UIE under the $D_{task,dist}$ and $D_{task,ind,dist}$ configurations, with average improvements of 2.9 and 3.2 percentage points over USM, respectively. This indicates that TRUE-UIE can adeptly generalize knowledge learned from relation extraction tasks to NER tasks within pre-training settings involving D_{dist} , despite the absence of annotated entity types.

Regarding zero-shot relation extraction, following USM, TRUE-UIE is trained on all available pre-training datasets, and benchmarked against GPT-3 175B (Brown et al., 2020) and DEEPSTRUCT 10B (Wang et al., 2022) on the Conll04 dataset. As shown in Table 3, despite having a smaller model size, TRUE-UIE not only surpasses robust zero-shot baselines such as GPT-3 and DEEPSTRUCTURE, but also demonstrates competitive performance compared to USM, which is of a comparable size. These findings robustly affirm the efficacy of the TRUE-UIE framework. Compared to multi-task models like USM, common task models manifest a superior capacity for generalization.

4.4 Experiments in the Few-shot Setting

In our few-shot transfer experiments, we followed the data preprocessing and experimental settings from previous studies (Lu et al., 2022; Lou et al., 2023). Table 4 shows the performance of 7 IE tasks in few-shot scenarios, with the average results from 1/5/10-shot experiments labeled as "Avg." TRUE-UIE*, representing the initial model without IE pre-training, is used as the baseline for discontinuous NER and Open IE tasks where UIE and USM are not applicable. The results indicate that TRUE-UIE outperforms both baseline models, achieving an average improvement of 6.29 and 1.17 on the first five datasets. This suggests a superior generalization ability over the other two baseline models. Moreover, TRUE-UIE surpasses its preliminary model, TRUE-UIE*, by an average score of 14.46 for the final three tasks. This demonstrates that TRUE-UIE is not only capable of expanding to more complex IE tasks but also effectively generalizes the knowledge gained during pretraining to novel tasks. These remarkable results stem from its architecture, which models IE tasks as a shared task using two universal relation extraction processes, maximizing knowledge sharing and robust scalability for vari-

Title	Model	1-Shot	5-Shot	10-Shot	Avg.
CoNLL03	UIE	57.53	75.32	79.12	70.66
	USM	71.11	83.25	84.58	79.65
	TRUE-UIE	73.56	84.78	85.66	81.33
CoNLL04	UIE	34.88	51.64	58.98	48.50
	USM	36.17	53.20	60.99	50.12
	TRUE-UIE	36.77	53.94	62.21	50.97
ACE05-Evt (trigger)	UIE	42.37	53.07	54.35	49.93
	USM	40.86	55.61	58.79	51.75
	TRUE-UIE	41.33	56.88	59.93	52.71
ACE05-Evt (argument)	UIE	14.56	31.20	35.19	26.98
	USM	19.01	36.69	42.48	32.73
	TRUE-UIE	19.64	37.10	43.55	33.43
Sentiment (16res)	UIE	23.04	42.67	53.28	39.66
	USM	30.81	52.06	58.29	47.05
	TRUE-UIE	32.03	54.02	60.12	48.72
Genia	TRUE-UIE*	6.10	29.33	33.44	22.96
	TRUE-UIE	37.34	55.54	57.97	50.28
Cadec _D	TRUE-UIE*	2.01	9.63	15.81	9.15
	TRUE-UIE	10.17	20.13	27.64	19.31
SAOKE	TRUE-UIE*	2.32	5.74	7.61	5.22
	TRUE-UIE	5.61	10.34	17.44	11.13

Table 4: Comparison of few-shot performance across various tasks. TRUE-UIE* indicates that the model is initialized from the original pre-trained language model.

ous tasks. Contrastingly, UIE’s need to learn varied schema structure languages leads to a large decoding search space and restricted knowledge sharing, presenting substantial learning challenges in low-resource settings. While USM reduces this search space via semantic matching, it fails to learn more universal relations, resulting in varied knowledge acquisition across tasks.

5 Conclusion

In this study, we’ve introduced an innovative approach called **TRUE-UIE**, which presents a unified framework for various information extraction (IE) tasks. By leveraging only two universal relations, namely IS and NEXT, we have established a consistent methodology across all IE tasks. This ensures that all components and definitions within the method remain uniform for different IE tasks, and can be applied to tasks such as discontinuous NER and open information extraction that are challenging for existing top-performing methods. The experimental results demonstrate that TRUE-UIE achieves state-of-the-art performance across 7 IE tasks and 16 datasets. It also showcases robust generalization capabilities in scenarios involving zero-shot and few-shot transfers. Notably, TRUE-UIE

offers both adaptable task scalability and the ability to seamlessly transfer pre-trained knowledge to novel tasks. We hope that TRUE-UIE can drive further development in the field of UIE to better explore the relevant knowledge between tasks.

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A More Dataset Details

A.1 Datasets for Evaluation

We carry out evaluations on 7 information extraction tasks, spanning 16 distinct datasets. Comprehensive statistics for each of these datasets are presented in Table 5. We follow the pre-processing steps and data split of previous works (Lu et al., 2022; Lou et al., 2023).

A.2 Datasets for Pretraining

Details regarding the pretraining datasets are outlined as follows:

- For D_{task} , all 60K samples are utilized.
- D_{dist} consists of 356K samples. From this, the Rebel dataset is narrowed down to the 230 most frequently occurring relation types, and 300K instances are randomly selected for pre-training, in accordance with Lou et al. (2023).
- D_{ind} contains 195K samples, drawn from several datasets: HotpotQA (Yang et al., 2018), Natural Questions (Kwiatkowski et al., 2019), NewsQA (Trischler et al., 2016), SQuAD (Rajpurkar et al., 2016), and TriviaQA (Joshi et al., 2017). For each instance, the selection is restricted to a maximum of 5 questions, and

Datasets	Ent	Rel/Pol	Evt	#Train	#Val	#Test
ACE04	7	-	-	6,202	745	812
ACE05-Ent	7	-	-	7,299	971	1,060
CoNLL03	4	-	-	14,041	3,250	3,453
Genia	5	-	-	16,692	1,854	1,854
Cadec	1	-	-	5,340	1,097	1,160
ACE05-Rel	7	6	-	10,051	2,420	2,050
CoNLL04	4	5	-	922	231	288
NYT	3	24	-	56,196	5,000	5,000
SciERC	6	7	-	1,861	275	551
ACE05-Evt	7	-	33	19,216	901	676
CASIE	21	-	5	11,189	1,778	3,208
14res	2	3	-	1,266	310	492
14lap	2	3	-	906	219	328
15res	2	3	-	605	148	322
16res	2	3	-	857	210	326
SAOKE	6	7	-	37,544	4,693	4,693

Table 5: The statistics for evaluation datasets

any samples where the combined text length exceeds 500 tokens are excluded.

- For the Chinese open information extraction (IE) dataset, Saoke, we deviate from the above datasets for pretraining. Instead, we assemble a large-scale distant supervision dataset by aligning Wikidata with the Chinese version of Wikipedia.

B Implementation Details

In all our experiments, the optimization of our model is performed using the Adam algorithm (Kingma and Ba, 2014). During the pretraining phase, we set the learning rate at 2×10^{-5} , the global batch size at 96, and run the process for 5 epochs. For the fine-tuning phase, we explore a variety of hyper-parameters, adjusting the learning rate within the range $\{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}, 4 \times 10^{-5}, 5 \times 10^{-5}\}$ and the batch size from among $\{8, 12, 16, 32, 64, 96\}$. With 3 random seeds, we select the optimal hyper-parameter configuration based on the performance on the development set. For multi-task learning, we choose the best checkpoint based on the average performance across all datasets. All experiments are carried out on NVIDIA A100 (80G) GPUs and repeated 3 times to reported the averaged F1 scores.

We evaluate the model using span-based offset Micro-F1 as the primary metric, with different criteria for different aspects of the information extraction task:

- **Entity:** An entity mention is deemed correct if both its offsets and type correspond to a reference entity.
- **Relation (Strict Match):** A relation is considered correct if its type matches and both the offsets and entity types of the related entity mentions are correct.
- **Relation (Triplet Match):** A relation is considered correct if its type matches, and the offsets of the subject and object are correct.
- **Event Trigger:** An event trigger is considered correct if its offsets and event type align with a reference trigger.
- **Event Argument:** An event argument is marked as correct if its offsets, role type, and event type match a reference argument mention.
- **Sentiment Triplet:** To consider a sentiment triplet correct, the offsets boundaries of both the aspect and the opinion span must be correct, and the sentiment polarity must also be accurate.

These criteria ensure a comprehensive evaluation of the model’s ability to correctly identify various elements of information extraction tasks.

C Ablation Study

In Table 6, we performed ablation studies on three components: token sequential dependency (Seq Dep) in span features, structure language prompt (SLP), and novel linking style for two universal relation extraction (TUR). We replaced span features with multiplicative attention and substituted SLP and TUR with USM’s naive prompt and linking style, excluding discontinuous NER and Open IE from the experiments since the naive method can not extend to these two tasks. Our conclusions:

- 1) Token sequential dependency is vital for all four IE tasks. Its removal led to a substantial performance decline, confirming its effectiveness.
- 2) Ablating SLP & TUR didn’t affect NER, as our prompt and linking style are similar to USM on the NER task. Other tasks showed declines, highlighting TRUE-UIE’s prompt and linking style’s effectiveness on IE tasks. The relatively noticeable performance decline in relation extraction and

event extraction demonstrates that this design effectively enhances the unification of learning objectives, allowing knowledge gained in NER to be shared across the relation extraction and event extraction tasks.

Task	Ent	Rel	Evt-Tri	Evt-Arg	Senti.
TRUE-UIE	96.89	68.91	73.12	58.33	81.73
w/o Seq Dep	95.26	67.52	72.79	57.34	80.91
w/o SLP & TUR	95.18	66.48	71.97	56.83	80.53

Table 6: Ablation study for TRUE-UIE on 4 tasks: entity recognition (CoNLL03), relation extraction (ACE-Rel), event extraction (ACE05-Evt), and sentiment analysis (16res).

D Limitations

The Structure Language Prompt might lead to performance decline in certain datasets where default entity types or coarse entity types are commonly used in many triplet schemes. This occurs as the same type of text, such as “people”, appears in different schemes, causing confusion. For instance, in Figure 1, “people” is used in both “work for” and “born in” relations, but an entity of the type “people” may not always be involved in both relations. If the model, post-training, represents “people” similarly across different schemes, it could lead to confusion, resulting in high recall but low precision. Our solution is to employ an attention mask strategy as following Figure 4, enabling the model to focus only on text within the scheme group. For example, the first “people” would only pay attention to “work for organization”, and the second “people” to “born in place”.

E Help from AI assistants

When necessary, we use ChatGPT or Copilot for guidance on how to write regular expressions, like the `tokenize_uni` function in `utils.py`.

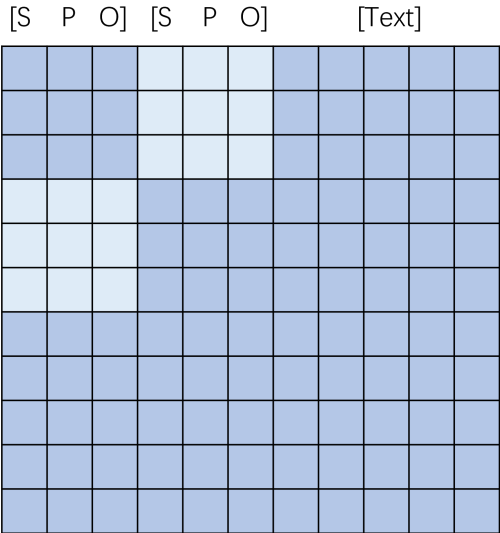


Figure 4: The figure illustrates TRUE-UIE’s attention mask approach for handling datasets with numerous duplicate entity/role types.